CURRICULUM SUMMARY: B.TECH.

CREDITS ALLOCATED TO DIFFERENT COURSE CATEGORIES IN DIFFERENT BRANCHES

Course Category /Branch	Artificial Intelligence	Automobile Engineering	Food Technology	Civil Engineering	Chemical Engineering	Computer Engineering	Electrical Engineering	Electronics Engineering	Mechanical Engineering	Petrochemical Engineering
PC	67	83.5	72.5	84.5	84	68	85.5	78	84.5	85.5
PE	18	15	15	15	15	15	15	15	15	15
BS	27	26	28.5	25	25	30	26	26	26	25
ESA	33	27.5	37	27.5	25	33	26.5	31	26.5	26.5
OE	6	6	6	6	6	6	6	6	6	6
HM	11	11	11	12	11	11	11	11	11	12
PSI	18	11	10	10	14	17	10	13	11	10

TOTAL CREDITS: 180

DESCRIPTION OF COURSE CATEGORIES

Course Categories	Required Credit
Basic Sciences (BS) (Courses such as Physics, Chemistry, Mathematics etc)	25-30
Engineering Sciences & Arts (ESA) (Foundation and applied engineering courses that are taught across many branches)	24-40
Humanities and Management (HM) (Language, Social science, & Management)	10-16
Programme Core (PC)	60-100
Project, Seminar, Internships (PSI)	10-18
Programme Electives (PE)	15-32
Open Electives (OE)	6-18
Audit Courses (AU) (Language, Social science, Management, value education, stress management, Environmental Studies etc.)	0

Notes:

1. A minimum of four of the Programme Elective/Open Elective courses, has to be taken from courses available on online platforms such as MOOCS, SWAYAM, NPTEL etc., as decided by the concerned Department/Centre.
2. Audit courses of II Year onwards can be off line courses or they can be taken from courses available on online platforms. One of the audit courses has to be in any of the foreign languages.
3. The Industrial Training/Internship can be undertaken during the vacations, however, the evaluation of the same shall be held in the 8th Semester.

Course Structure: B.Tech. (Valid for students admitted from year 2023 onwards)

First Year-All Branches (Section A1A, A1B & A1C)

Semester 1:

S.No.	Couse Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Couse-Work	Mid-Sem	End-Sem	
1	BS	AMS1112	Applied Mathematics-I	3	1	0	4	15	25	60	100
2	BS	APS1112	Applied Physics	3	1	0	4	15	25	60	100
3	ESA	ELA1112	Principles of Electronics Engineering	2	1	0	3	15	25	60	100
4	ESA	MEA1122	Engineering Mechanics	2	1	0	3	15	25	60	100
5	HM	EZH1112	English	2	1	0	3	15	25	60	100
6	BS	APS1912	Applied Physics Lab	0	0	3	1.5	60		40	100
7	ESA	MEA1922	Engineering Workshop	0	1	2	2	60		40	100
8	HM	EZH1912	Communication Skills Lab	0	1	2	2	60		40	100

TOTAL CREDITS: 22.5

Semester 2:

S.No.	Couse Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Couse-Work	Mid-Sem	End-Sem	
1	BS	AMS1122	Applied Mathematics-II	3	1	0	4	15	25	60	100
2	BS	ACS1112	Applied Chemistry	3	1	0	4	15	25	60	100
3	ESA	EEA1112	Principles of Electrical Engineering	2	1	0	3	15	25	60	100
4	AU	CEU1112	Environmental Studies (Audit Course)	3	0	0	0	15	25	60	100
5	ESA	MEA1112	Thermal Sciences	3	1	0	4	15	25	60	100
6	BS	ACS1912	Applied Chemistry Lab	0	0	3	1.5	60		40	100
7	ESA	COA1912	Computer Programming Lab	0	1	2	2	60		40	100
8	ESA	MEA1912	Engineering Graphics Lab	1	0	2	2	60		40	100

TOTAL CREDITS: 20.5

First Year-All Branches (Section A1D, A1E & A1F)

Semester 1:

S.No.	Couse Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Couse-Work	Mid-Sem	End-Sem	
1	BS	AMS1112	Applied Mathematics-II	3	1	0	4	15	25	60	100
2	BS	ACS1112	Applied Chemistry	3	1	0	4	15	25	60	100
3	ESA	EEA1112	Principles of Electrical Engineering	2	1	0	3	15	25	60	100
4	AU	CEU1112	Environmental Studies (Audit Course)	3	0	0	0	15	25	60	100
5	ESA	MEA1112	Thermal Sciences	3	1	0	4	15	25	60	100
6	BS	ACS1912	Applied Chemistry Lab	0	0	3	1.5	60		40	100
7	ESA	COA1912	Computer Programming Lab	0	1	2	2	60		40	100
8	ESA	MEA1912	Engineering Graphics Lab	1	0	2	2	60		40	100

TOTAL CREDITS: 20.5

Semester 2:

S.No.	Couse Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Couse-Work	Mid-Sem	End-Sem	
1	BS	AMS1122	Applied Mathematics-I	3	1	0	4	15	25	60	100
2	BS	APS1112	Applied Physics	3	1	0	4	15	25	60	100
3	ESA	ELA1112	Principles of Electronics Engineering	2	1	0	3	15	25	60	100
4	ESA	MEA1122	Engineering Mechanics	2	1	0	3	15	25	60	100
5	HM	EZH1112	English	2	1	0	3	15	25	60	100
6	BS	APS1912	Applied Physics Lab	0	0	3	1.5	60		40	100
7	ESA	MEA1922	Engineering Workshop	0	1	2	2	60		40	100
8	HM	EZH1912	Communication Skills Lab	0	1	2	2	60		40	100

TOTAL CREDITS: 22.5

B.TECH. (CHEMICAL ENGINEERING)

Semester 3:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	PC	CHC2012	Basic Principles of Chemical Engineering	3	1	-	4	15	25	60	100
2	PC	CHC2032	Fluid Mechanics	3	1	-	4	15	25	60	100
3	PC	CHC2042	Fluid Particle Operations	3	1	-	4	15	25	60	100
4	PC	CHC2052	Chemical Engineering Thermodynamics	3	1	0	4	15	25	60	100
5	PC	CHC2062	Heat Transfer Operations	3	1	0	4	15	25	60	100
6	PC	CHC2072	Numerical Methods in Chemical Engg.	3	1	0	4	15	25	60	100
7	PC	CHC2902	Computer Applications Lab			3	1.5	60	--	40	100

TOTAL CREDITS: 25.5

Semester 4:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	BS	ACS2022	Engineering Chemistry	3	0	0	3	15	25	60	100
2	PC	CHC2002	Process Instrumentation	3	0	0	3	15	25	60	100
3	PC	CHC2022	Chemical Reaction Engineering	3	1	0	4	15	25	60	100
4	PC	CHC2082	Mass Transfer Operations	3	1	0	4	15	25	60	100
5	PC	CHC2092	Equilibrium Stage Processes	3	1	0	4	15	25	60	100
6	PC	CHC2922	Fluid Mechanics Lab	-		3	1.5	60	--	40	100
7	PC	CHC2932	Mechanical Operations Lab	-		3	1.5	60	--	40	100
8	PC	CHC2942	Heat Transfer lab	0	0	3	1.5	60		40	100

TOTAL CREDITS: 22.5

B.TECH. (CHEMICAL ENGINEERING)

Semester 5:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	ESA	ACA3112	Engineering Materials	3	0	0	3	15	25	60	100
2	BS	AMS3412	Advanced Engineering Mathematics	3	0	0	3	15	25	60	100
3	ESA	CHA3012	Biology for Engineers	3	0	0	3	15	25	60	100
4	PC	CHC3122	Process Equipment Design	3	0	0	3	15	25	60	100
5	PC	CHC3912	Computer Programming Lab	0	0	3	1.5	60		40	100
6	PC	CHC3952	Reaction Engg. & Thermodynamics Lab			3	1.5	60		40	100
7	PC	CHC3962	Mass Transfer Lab	0	0	3	1.5	60		40	100
8	PSI	CHP3812	Seminar and Report Writing	0	2	0	2	100			100
9	HM	MEH3452	Engineering Economy & Management	3	0	0	3	15	25	60	100

TOTAL CREDITS: 21.5

Semester 6:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	PC	CHC3132	Chemical Process Industries	3	1	0	4	15	25	60	100
2	PC	CHC3162	Energy Engineering and Management	3	0	0	3	15	25	60	100
3	PC	CHC3182	Transport Phenomena	3	1	-	4	15	25	60	100
4	PC	CHC3192	Plant Design and Economics	3	1		4	15	25	60	100
5	PC	CHC3922	CAD/Simulation Lab	0	0	3	1.5	60		40	100
6	PC	CHC3972	Chemical Engineering Design & Drawing	0	0	2	1	15	25	60	100
7	PSI	CHP3932	Plant Design Project A	0	3	0	3	60		40	100
8	PE		Programme Elective-1				3	15	25	60	100
9	PE		Programme Elective-2				3	15	25	60	100

TOTAL CREDITS: 26.5

B.TECH. (CHEMICAL ENGINEERING)

Semester 7:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	PC	CHC4112	Industrial Pollution Control & Waste Minimization	3	0	0	3	15	25	60	100
2	PC	CHC4122	Process Dynamics and Control	3	1	0	4	15	25	60	100
3	PC	CHC4232	Process Safety and Risk Analysis	3	0	0	3	15	25	60	100
4	PC	CHC4922	Process Dynamics and Control lab	0	0	3	1.5	15	25	60	100
5	PC	CHC4932	Technology Lab	0	0	3	1.5	60		40	100
6	PSI	CHP4982	Plant Design Project B	0	3	0	3	60		40	100
7	PSI	CHP4992	Project (Experimental/Computational)	0	3	0	3	60		40	100
8	OE		Open Elective-1				3	15	25	60	100
9	HM		Elective Course under Humanities category				3	15	25	60	100
10	AU		Audit Course-2				0	15	25	60	100

TOTAL CREDITS: 25

Semester 8:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	PC	CHC4952	Industrial Lectures	0	1	0	1	15	25	60	100
2	PSI	CHP4802	Industrial Training/Internship	0	2	0	2			100	100
3	PSI	CHP4812	Social Internship	0	2	0	1			100	100
4	PE		Programme Elective-3				3	15	25	60	100
5	PE		Programme Elective-4				3	15	25	60	100
6	PE		Programme Elective-5				3	15	25	60	100
7	OE		Open Elective-2				3	15	25	60	100
8	AU		Audit Course-3				0	15	25	60	100

TOTAL CREDITS: 16